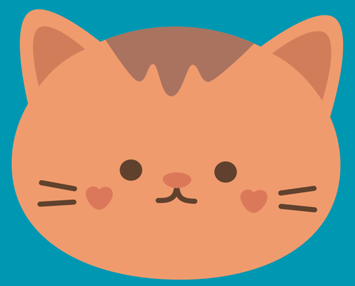
Q-Volution 2026: Track C

Schrödinger's Wildcats

LCU for Quantum Harmonic Oscillator

Members of Group

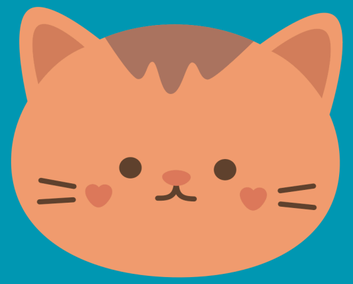
Kristy



Afra

Vivian

Sara



Anagha

Introduction to the Problem

Differential equations are the primary means of modeling the dynamics of physical, engineering, and computational systems. From the motion of mechanical systems to the propagation of electromagnetic fields, many physical processes are governed by linear differential equations. The efficient solution of these equations is one of the main problems addressed by scientific computing.

Quantum computing provides a new paradigm for the solution of these problems. Quantum algorithms represent the entire solution vector as the amplitudes of the quantum state. This new paradigm enables new computational strategies.

A new framework for the solution of these problems is provided in the paper “A Quantum Algorithm for Solving Linear Differential Equations: Theory and Experiment.”

This paper presents the use of the truncated Taylor series expansion along with the Linear Combination of Unitaries (LCU) technique.

In this work, we implement this algorithmic framework and apply it to a fundamental physical system, analyzing both its numerical accuracy and its quantum resource requirements.

Algorithm Overview

1) Taylor Expansion & Quantum State Expression

Use Taylor Expansion to get rid of matrix exponents. Convert the eq. to a quantum state (Bra-Ket)

2) Create Registers

- Register 1 (ancilla) acts as a “switch” to encode/decode x(0) or b
- Register 2 (ancilla) encodes iterations of Taylor Expansion
- Register 3 (work) encodes state of solution

3) Encode Starting States

Apply V to Register 1. Then perform controlled operations with Register 1 being the control and Register 2 & 3 being the target, applying V_S1 and V_S2 to 2 and U_x and U_b to 3.

4) Entanglement Generation

Do a series of controlled operations with Register 2 being the control and Register 3 being the target, effectively applying the correct operator to the state for each iteration of the Taylor Expansion with the correct Taylor weight. These two registers are now entangled.

5) Decode States

Perform controlled operations with Register 1 being the control and Register 2 being the target, applying the Hermitian to the V_S1 and V_S2 operators (W_S1 and W_S2). Then, apply the Hermitian to the V operator “W” to Register 1.

6) Measurement

Measure the system. The states where Register 1 and 2 equal |0> will have the correct output for the chosen time and iteration number!

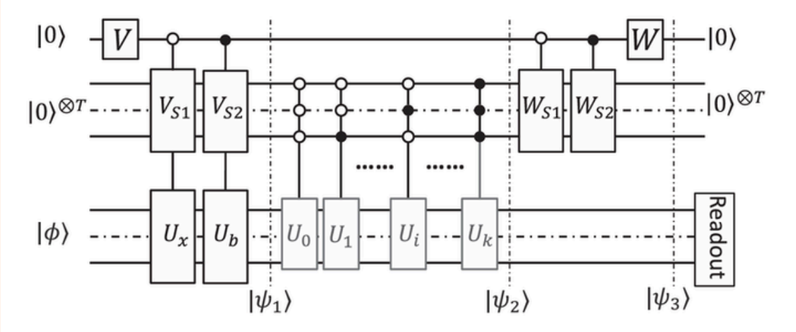


Figure of a circuit executing a generalized version of this algorithm. Credit to Tao Xin et. al. (doi.org/10.1103/PhysRevA.101.032307)

Applied Theory

$$y'' + \omega^2 y = 0\ ,\ y(0) = 1,\ y'(0) = 1,\ \omega = 1$$

Given Equation

We can break up the given second order differential into a system of first order differential equations

$$\frac{d}{dt}\begin{bmatrix} y \\ y' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} y \\ y' \end{bmatrix}$$

From here, it is clear that our matrix M is proportional to the Pauli Y gate.

$$M^m = (i\sigma_y)^m$$
$$\Rightarrow m = 0:\ I$$
$$\Rightarrow m = 1: M = iY$$
$$\Rightarrow m = 2:\ M^2 = -I$$
$$\Rightarrow m = 3:\ M^3 = -iY$$
$$\Rightarrow m = 4:\ M^4 = I$$

Because the modulus of M is 1 and the coefficient b is 0, the sum of our coefficients C_m and D_n simplify to:

$$\sum_{m=0}^{k} C(t) = 1 + t^1 + \frac{t^2}{2} + \frac{t^3}{6} + \frac{t^4}{24} + \frac{t^5}{120} + \ldots + \frac{t^k}{k!} = e^t$$

$$\sum_{n=1}^{k} D_n = 0$$

This result makes our operator V equal to the Pauli Z gate.

$$V(t) = e^{-t/2}\begin{bmatrix} e^{t/2} & 0 \\ 0 & -e^{t/2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Due to the initial conditions, the starting state must be equally in both y and y'. This is achieved through the use of the H gate.

$$x(0) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \Rightarrow normalize \Rightarrow |x(0> = \frac{1}{\sqrt{2}}(|0> + |1>)$$

The final operators to find are V_S1, W and W_S1 (W_S2 = 0). Because V is the Z gate, W would also be the Z gate. For the other two, the requirements are that the first row be a list of C_m values, and that W_S1 is the Hermitian of V_S1. This criteria will be accomplished through code implementation of the Gram-Schmidt method.

After applying the algorithm using our found operators, the final state in Register 3 will give us information about y and y'. Using the following formulas, we will be able to extract the kinetic and potential energies of our system.

Potential Energy: $PE = \frac{1}{2}ky^2$

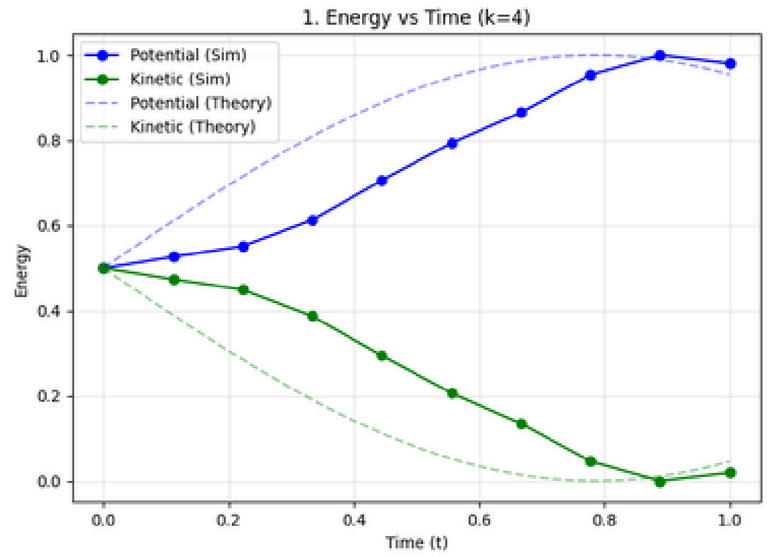
Kinetic Energy: $KE = \frac{1}{2}m(y')^2$

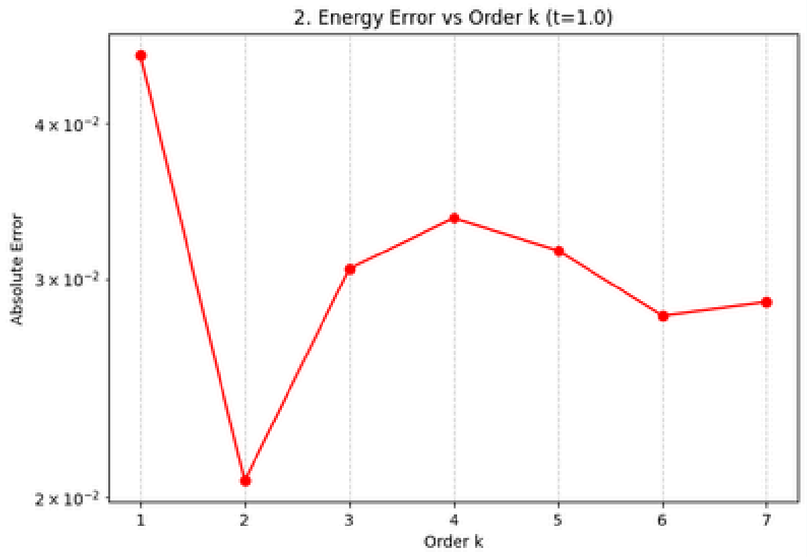
By repeating this process for different time values, we can see how a quantum harmonic oscillator's potential and kinetic energies fluxuate over time.

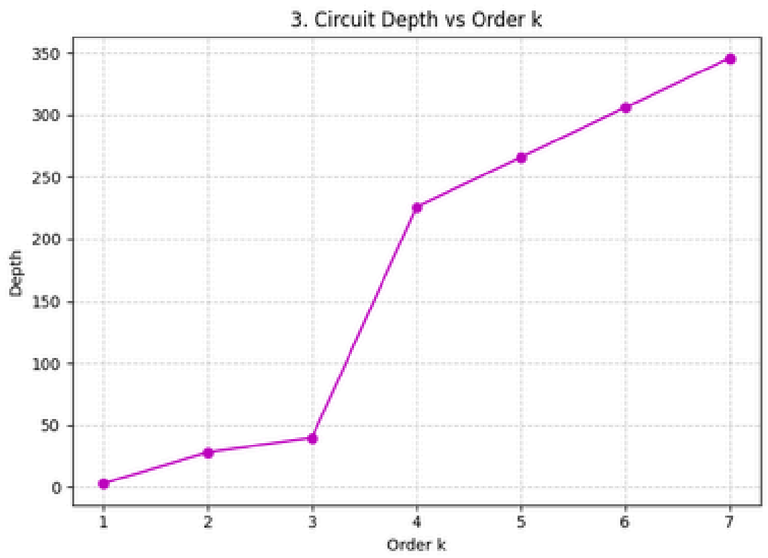
Analysis

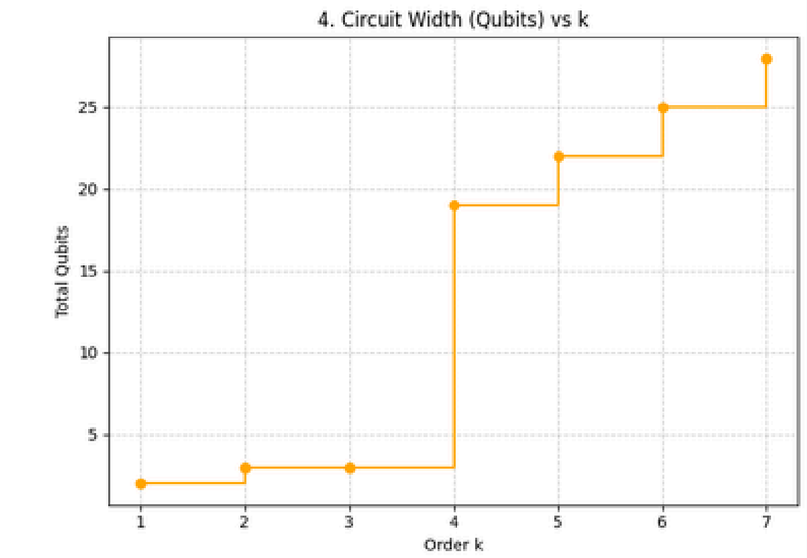
Our Energy vs Time graph matches our predictions. At t = 0, the given initial conditions y(0) = 1 and y'(0) = 1 indicate that the PE = ½ $|y|^2$ = 0.5 and KE = ½ $|dy/dx|^2$ =0.5. As time evolves these inputs diverge in sine and cosine solutions.

Notice how circuit depth increases with the order k, so even though increasing k increases the precision of the alogirithm, there is a computational advantage to truncating the Taylor expansion.







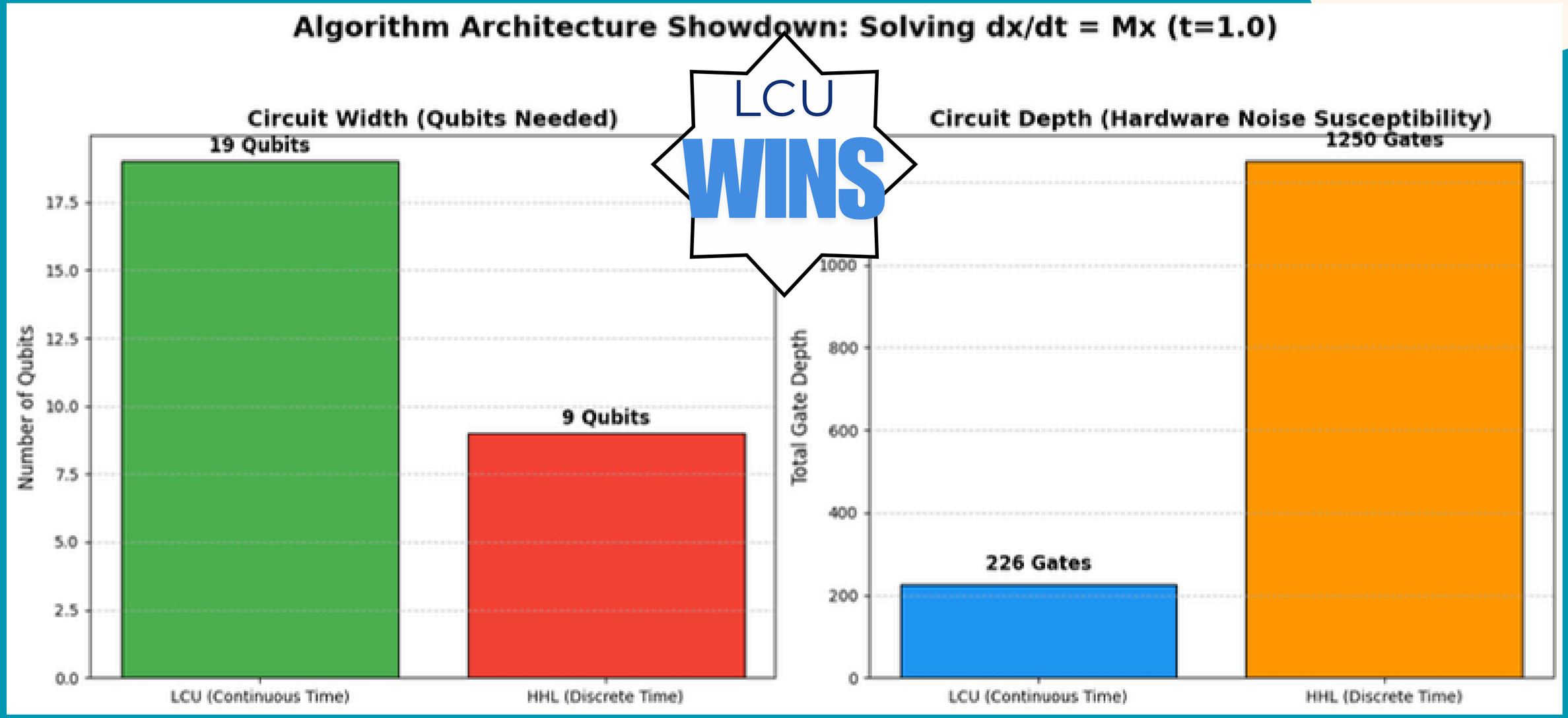


While we have an unexpected dip at k = 2 and peak at k = 4, our general trend confirms that as k increases the energy error decreases (i.e. precision increases)

Another optimization constraint shows that as k increases, circuit width (aka number of qubits necessary) increases. This too supports truncating the series.

LCU vs HHL

Harrow, Hassidim, and Lloyd (2009)



Utility of our Project

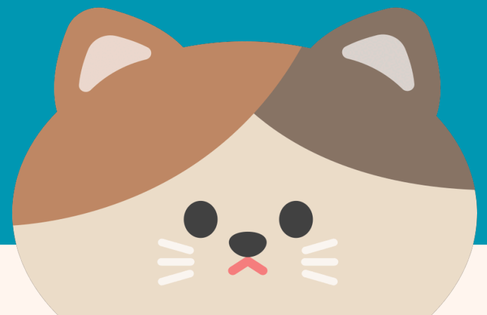
Our approach can be applied to solve any linear system of the form x' = Ax

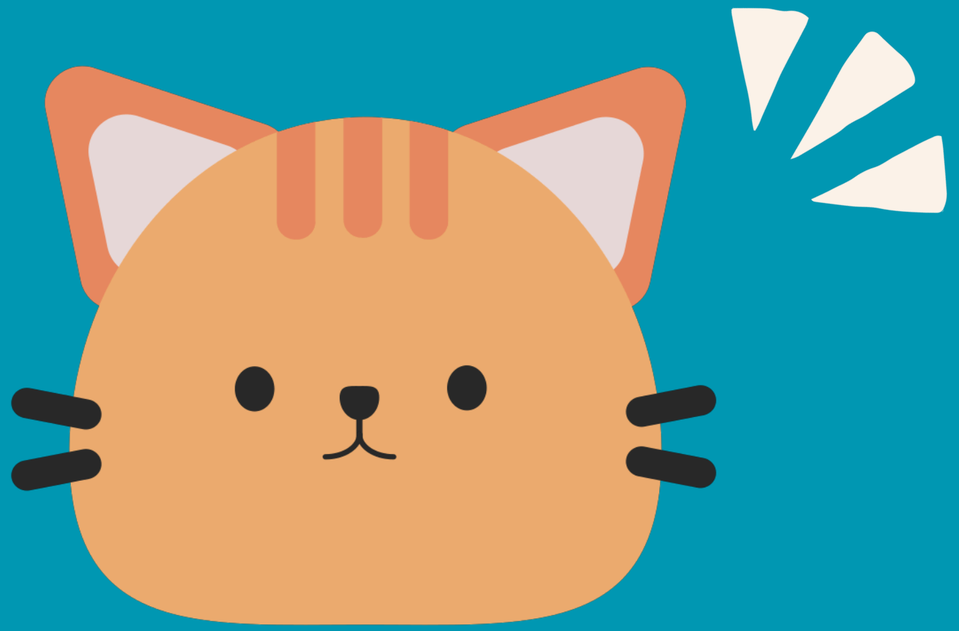
Real-life applications:

- Quantum Computer hardware: superconducting Qubits and microwave resonators behave like a simple harmonic oscillator
- Quantum Chemistry: Simulating molecular vibrations
- Quantum Optics: Quantizing the electromagnetic field

Further Areas of Research

- Extend to non-homogeneous linear ODEs to simulate driven systems where external forces are present (i.e., nonzero b vector)
- Scale to higher-dimensional systems, such as coupled oscillators or larger sparse matrices
- Implementing our quantum circuit on real hardware to analyse performance





For your attention we say:

Thank You!